Senthil Rajasekaran

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EDUCATION

Rice University, Houston, TX — Fall 2019 - present
PhD in Computer Science, GPA 3.88/4.00
Advisor: **Dr. Moshe Vardi**

University of Oxford, Oxford, UK — 9/2016 - 9/2017
MSc in Mathematics and the Foundations of Computer Science

Tulane University, New Orleans, LA — May 2015
BS in Mathematics, BS in Economics
Magna Cum Laude with departmental honors in Mathematics

SELECTED WORKS

Nash Equilibria in Finite-Horizon Multiagent Concurrent Games.
Senthil Rajasekaran and Moshe Vardi
In Proceedings of the 20th International Conference on Autonomous Agents and MultiAgent Systems (**AAMAS '21**).
This paper provides a **new PSPACE-complete complexity result** for the problem of finding Nash Equilibria in finite-horizon multiagent games where each agent has a behavioral goal, one that depends on the history of the play, expressed as a deterministic finite automaton.

Verification and Realizability in Finite-Horizon Multiagent Systems
Senthil Rajasekaran and Moshe Vardi
In Proceedings of the 19th International Conference on Principles of Knowledge Representation and Reasoning (**KR '22**)
Here, we extend the results of the previous paper for new complexity results for both different types of automaton representation of the agent goals and with respect to the problem of checking equilibria. In doing so, **we also solve the previously open problem of determining the complexity of the nondeterministic finite automaton realizability problem.**

Multi-Agent Systems with Quantitative Satisficing Goals
Senthil Rajasekaran, Suguman Bansal, and Moshe Vardi
In Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (**IJCAI '23**)
In this paper we show how to leverage the automata-based techniques developed in the above two papers to **introduce a new way to specify goals in quantitative games with beneficial game-theoretic properties** and show how to develop efficient automata-based algorithms to solve quantitative games in which agents are given these types of goals **providing one of the first algorithms for finding equilibria in concurrent multiagent games with general quantitative goals.** Press: https://csweb.rice.edu/news/quantitative-goal-approach-game-theory-problem-could-be-important-building-block

INTERNSHIP

Nokia Bell Labs, Murray Hill, NJ — 6/2022 – 09/2022
Advisor: **Kedar Namjoshi**
In this internship, Kedar and I worked to build a proof assistant in OCaml to help users write provably correct distributed protocols. The work we did over the summer is currently being prepared for a conference paper submission.

SKILLS

I have strong mathematical skills with a special interest in **game theory, automata theory, and logic.** With respect to programming, I'm experienced with **Python and functional programming**.

SELECTED SERVICE AND AWARDS

- Awarded travel grants to present all three papers in "Selected Works"
- Awarded a special travel grant to represent Rice University in an exchange with Edinburgh University
- Programming Director for Rice Graduate Student Alliance (GSA) (2022-2023)
- Received two invitations to the Simons Institute to discuss my work in an international forum